Towards Foundational Models For Single-Chip Radar

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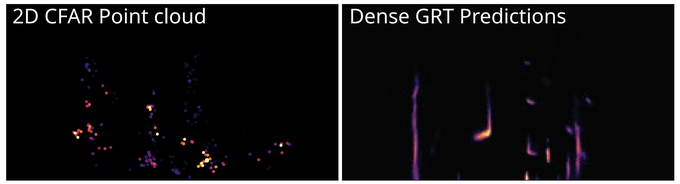
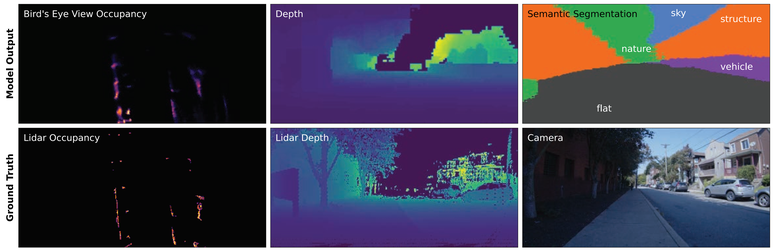
mmWave radars are compact, inexpensive, and durable sensors that are robust to occlusions, weather and darkness. Radar data is generally processed into point clouds using classical signal processing techniques; however, these point clouds have low density and poor quality, especially with inexpensive single-chip radars.

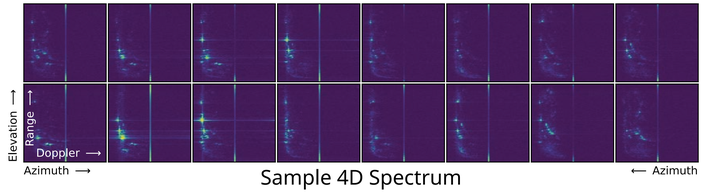


While ML-based approaches using raw spectrum are a promising alternative, existing work is largely limited by the lack of data. By scaling data collection, we train GRT: the **Generalizable Radar Transformer**, and show that with enough training data, 3D outputs of surprising quality can be extracted from low-resolution single-chip radars using a generalizable and fine-tunable model.



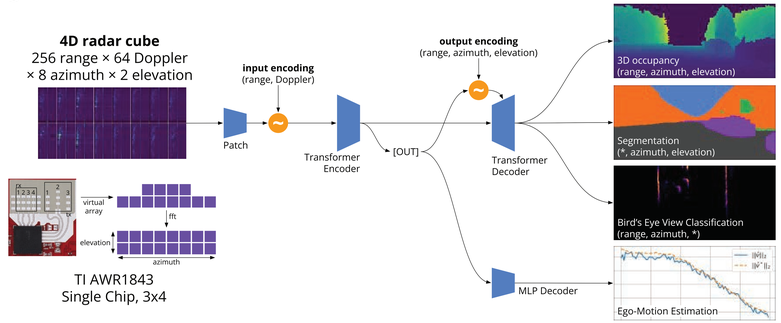
GRT model outputs, using only unfiltered 4D spectrum from a low-resolution single-chip mmWave radar as input.

Dataset: Using our *red-rover* data collection rig, we collect over 29 hours of radar-lidar-camera data (≈1M samples @ 10Hz, including radar I/Q time-signal) split between indoors, outdoors (handheld), and on bike — an order-of-magnitude increase over prior public datasets.

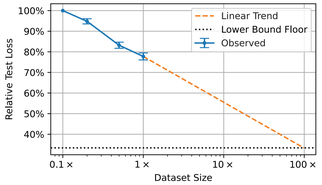


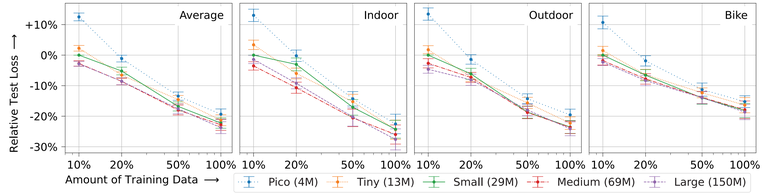
Sample 4D Spectrum

Method: GRT is an encoder-decoder transformer model with multi-dimensional sinusoidal positional encodings. We train a GRT base model using a (polar) 3D occupancy objective with Lidar as ground truth, which we then fine-tune for downstream tasks.



Key Findings: GRT shows logarithmic scaling (20% per 10× data) over dataset size, but no substantial scaling over model size, suggesting that much more data is required. Projecting this trend forward, we estimate 100× more data is required.





We also run a range of ablations on key architectural parameters; see the paper for details!

use spectrum instead of point clouds	use 4D (with Doppler) instead of 3D spectrum	train separate models for each setting
–31%	**–23%**	**+3-6%**

What Next? We believe the radar research community has a pressing need for standardized, high-quality, fully open source tooling, datasets, and libraries for radar spectrum. GRT is just the first step — please get in touch!

RadarML
a modular ecosystem for learning on radar spectrum
radarml.github.io